A 9 Pulley drive

Knowns

$dia_{pulley} = 3''$

1500 RPM

5 lbf Static

Steel Pulley

$\mu = .5$

$S_y = 3.25$ MPa @ 20% Strain

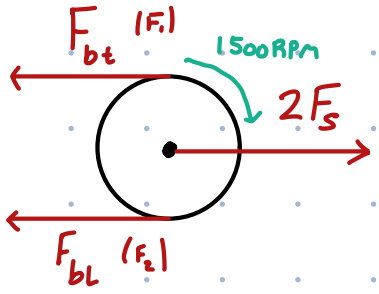
Center dist. = 5in

Unknowns

F_{bt}, F_{bL}

dia_{belt}, Cir_{belt}

FBD





$P = T \cdot \omega \qquad P = r(F_1 - F_2)\omega \qquad \frac{F_1}{F_2} = e^{\mu\theta}$

$\sum F_x = F_T + F_t = 2F$

$F_t = 2F - F_t \qquad F_t = 2F - F_T \longrightarrow \frac{F_t}{2F_s - F_t} = e^{\mu\theta}$

$\frac{F_t}{e^{\mu\theta}} = 2F_s - F_t \rightarrow \frac{1}{e^{\mu\theta}} = \frac{2F_s}{F_t} - \frac{F_t}{F_t} \rightarrow \frac{1}{e^{\mu\theta}} + 1 = \frac{2F_s}{F_t}$ (the $\frac{F_t}{F_t}$ term cancels to 1)

$\frac{e^{\mu\theta}}{1+e^{\mu\theta}} = \frac{2F_s}{F_t} \rightarrow F_t = 2F_s\left(\frac{e^{\mu\theta}}{1+e^{\mu\theta}}\right)$

$F_t = 2F_s - F_t$

$F_t = 2(5)\left[\frac{e^{(.5)(\pi)}}{(1+e^{(.5)(\pi)}}\right] = 8.279$ lb

$A = \frac{F}{\sigma} \qquad A = \frac{\pi d^2}{4}$

$\frac{\pi d^2}{4} = \frac{F}{\sigma}$ ↓ F.S

$d = \sqrt{\frac{(4)(F)(F.S)}{(\pi)(\sigma)}}$

$$F_L = 2(5) - 8.279 = 1.721 \text{ lb}$$

$$Cir = 2C + \frac{\pi}{2}(D+d) + \frac{(D-d)^2}{4C} + \pi\phi$$

$$d = \sqrt{\frac{(4)(8.279)(3)}{(\pi)(471)}} = .259 \text{ in}$$

↳ 3/8" belt used

$$Cir_{out} = 2(5) + \frac{\pi}{2}(3+3) + \frac{(3.3)^2}{4C} + \pi(.259) = 20.238 \text{ in}$$

↳ 20 1/4 in

McMaster-Carr ID: 3044K608 - 1835T909

Knowns:

$S_y = 36$ ksi

SF = 3

$F_t = 8.279$ lb

$F_L = 1.721$ lb

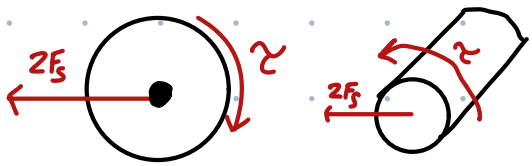
$dia_{Pulley} = 3$ in

Unknowns:

τ

dia_{Shaft}

FBD



$$\frac{Tc}{J} = \vec{\tau} = S_y \quad \left.\right\} + F.S$$

$$J = \frac{\pi d^4}{32}$$

$$C = r$$

$$(S_y) = \frac{(32)(T)(\frac{d}{2})(F.S)}{(\pi)(d)^4}$$

$$\downarrow$$

$$d = \sqrt[3]{\frac{(16)(T)(F.S)}{(\pi)(S_y)}}$$

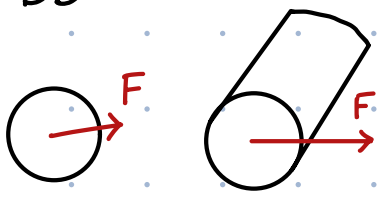
$$T = r(F_1 - F_2)$$

$$T = \left(\frac{3}{2}\right)\left(8.279 - 1.721\right) = 9.837 \text{ in}\cdot\text{lb}$$
(3/2 in; 1 lb)

$$d = \sqrt[3]{\frac{(16)(9.837)(3)}{(\pi)(36{,}000)}} = .161 \text{ in}$$

Knowns:	Unknowns	Assumptions
RPM = 1500	linear vel.	L = .25 in
Shaft dia = .161 in	Pressure	
F = 5 lb	PV	

FBD



$$P = \frac{F}{L \cdot d} \qquad V = RPM \cdot \pi \cdot d$$

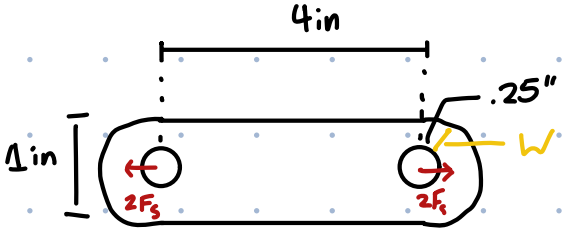
$$P = \frac{5}{(.25)(.161)} = 198.758 \text{ psi}$$

$$V = (1500)(\pi)(.161) = 758.695 \frac{\text{in}}{\text{min}}$$ (RPM)

$$758.695 \text{ in/min} = 63.225 \text{ ft/min}$$

$$PV = 198.758 \cdot 63.225 = 12{,}566.386 \frac{\text{psi}\cdot\text{ft}}{\text{min}}$$

Based on textbook → Porous Bronze or PTFE/RTFE



I Chose 6061 T6 Al as material

$$S_y = \frac{F}{A} \rightarrow T = \frac{F(F.S)}{w \cdot S_y}$$

$$A = w \cdot T$$

$$T = \frac{(20)(3)}{(.375)(40{,}000)}$$

$T = 4 \times 10^{-3}$ in
(very thin)

Thickness of .25 in, for ease of handeling.